

Extraction of Effective Temperatures from Tsallis Fits to Transverse-Momentum Spectra of Direct Photons and Heavy Quarkonia in High-Energy Collisions

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Abstract:

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I. RESULTS AND DISCUSSION

The Boltzmann distribution characterizes systems in equilibrium, whereas the Tsallis distribution describes systems away from equilibrium. The entropic index q quantifies the degree of departure from equilibrium and serves as an effective parameter that remains valid across the entire temporal and spatial domains as well as over the full transverse-momentum spectrum.

The effective temperature extracted from the Tsallis fits lies in the range 0.11–0.14 GeV and exhibits almost no dependence on centrality, which precisely reflects the complexity of nonprompt-photon production. Nonprompt photons originate almost entirely from contributions spanning the early to late stages of the quark–gluon plasma (QGP): the QGP evolves from a phase-transition temperature of ~ 0.17 GeV in the early stage down to a freeze-out temperature of ~ 0.10 GeV at the late stage, and the blue shift arising during this evolution drives the apparent effective temperature to higher values.

A central feature of the nonprompt direct-photon spectra [1] is that they are not described by a single exponential: the local inverse slope rises with transverse momentum, and the data were accordingly characterized by two separate exponential fits, yielding $T_{\text{eff}} \approx 0.26$ GeV/ c in the range $0.8 < p_T < 1.9$ GeV/ c and $T_{\text{eff}} \approx 0.38$ GeV/ c in $2 < p_T < 4$ GeV/ c . The present analysis shows that this evolving slope is captured naturally, and without piecewise treatment, by a single Tsallis form fitted over the full momentum range.

The equivalence follows directly from the structure of the Tsallis function. For $dN/dp_T \propto p_T [1 + (q - 1)p_T/T]^q$, the local inverse slope is not constant but a smooth, monotonically increasing function of p_T ,

$$T_{\text{eff}}(p_T) \equiv - \left(\frac{d \ln(dN/dp_T)}{dp_T} \right)^{-1}, \quad (1)$$

which reduces to the constant T only in the Boltzmann limit $q \rightarrow 1$. A value $q > 1$ therefore encodes precisely the spectral hardening that the two-exponential description represents through a discrete jump in T_{eff} . The single parameter q thus subsumes the two independent slopes of the piecewise fit.

This is borne out quantitatively. Evaluating $T_{\text{eff}}(p_T)$ for the 0%–20% best-fit parameters ($q = 1.084$, $T = 0.128$ GeV) gives $T_{\text{eff}} \approx 0.26$ GeV/ c near $p_T = 1.3$ GeV/ c and $T_{\text{eff}} \approx 0.40$ GeV/ c near $p_T = 3$ GeV/ c , in close

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agreement with the two inverse slopes extracted independently [1] for the corresponding p_T intervals. The two regimes are recovered not as separate fits but as the low- and high- p_T limits of one continuous curve.

The Tsallis description is therefore both more economical and more physically suggestive. It replaces two phenomenological slopes by a single functional form with one shape parameter q , and it admits a transparent interpretation within superstatistics: a Tsallis spectrum arises from a superposition of Boltzmann emitters with a distribution of inverse temperatures β , with $q - 1 \simeq \text{Var}(\beta)/\langle\beta\rangle^2$. The hardening described piecewise by two slopes is thus reframed as a single spectrum emitted by a system spanning a range of temperatures—a picture consistent with photon emission integrated over the cooling history of the expanding fireball.

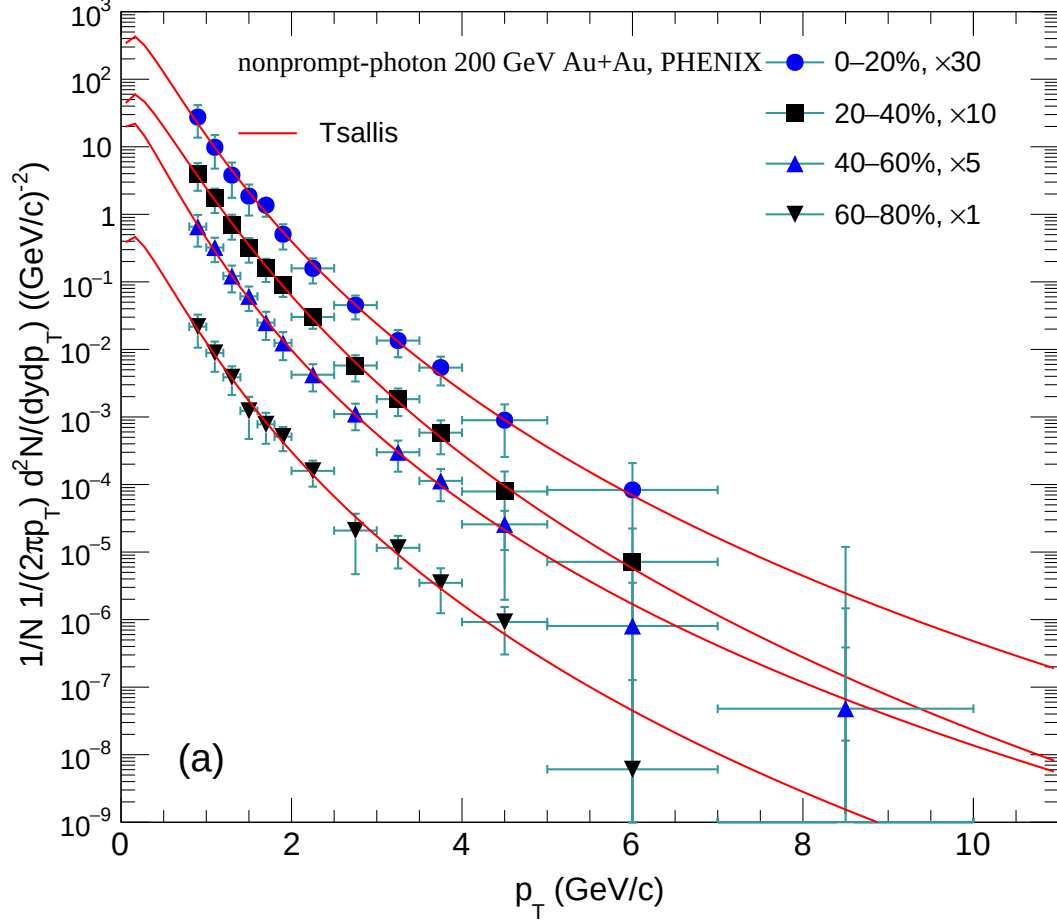


Figure 1a. Nonprompt direct-photon spectra in Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV for the four centrality classes, with the Tsallis fits (curves) evaluated at the best-fit (q, T, Nor) values.

[1] N. J. Abdulameer *et al.* (PHENIX Collaboration), “Nonprompt direct-photon production in Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV,” *Physical Review C* 109, no. 4 (2024): 044912, <https://doi.org/10.1103/PhysRevC.109.044912>.

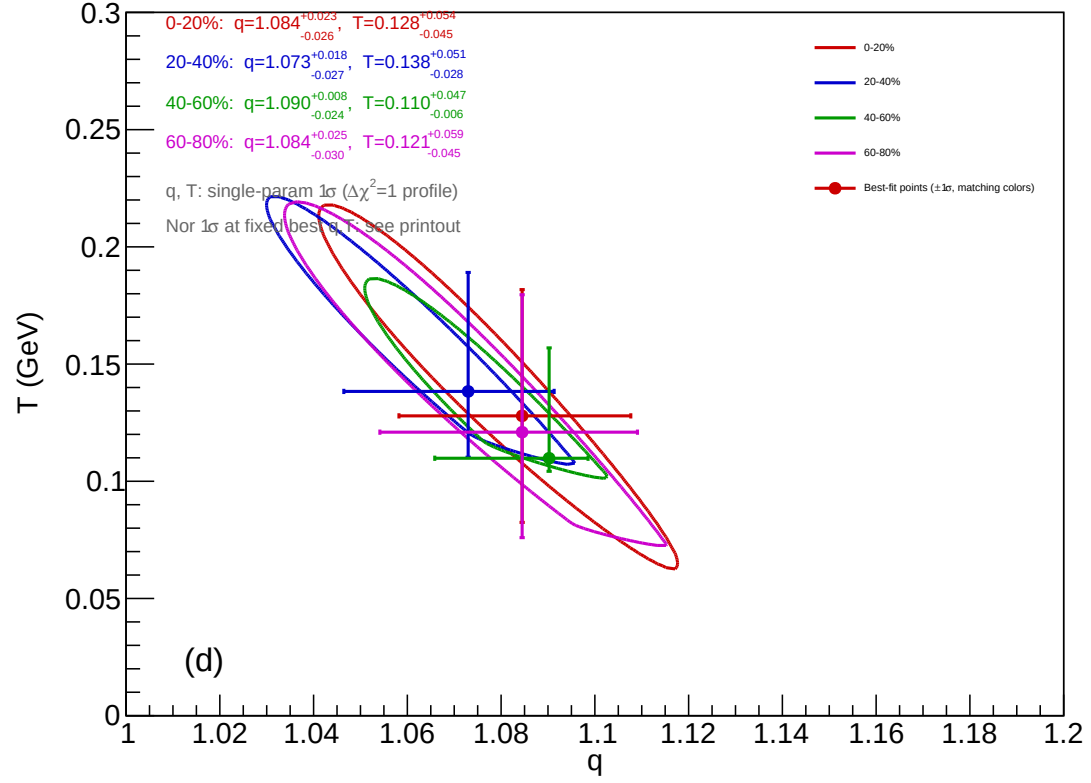


Figure 1d. Tsallis-fit confidence regions in the (q, T) plane for four centrality classes. Closed curves show the 68.3% joint confidence region ($\Delta\chi^2 = 2.30$); filled circles mark the best fits, with bars giving single-parameter 1σ errors ($\Delta\chi^2 = 1$).